AUDIT OF INPUT INTEGRITY & REASONING TRACES

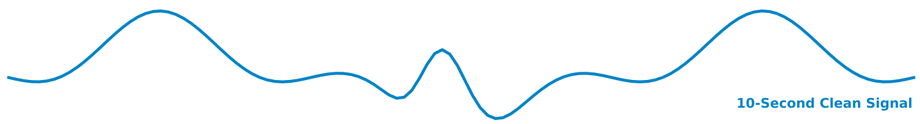
Verification of Signal Provenance, Attention Attribution, and Discovery of the Ungrounded Reasoning Limitation

SPOKEN SCRIPT: "***We have also examined the integrity of our inputs and reasoning traces, identifying limitations that must be addressed before drawing stronger conclusions.***"

PART A: INPUT INTEGRITY & SENSOR PROVENANCE AUDIT

1. Signal Preprocessing & Bandpass Verification

Harmonized 12-Lead Tensor (100 Hz, 0.5 - 40 Hz Butterworth filtering)



2. Controlled Physiological Masking (No Inpainting)

Active vs. Zeroed Physical Leads in Reduced Telemetry (e.g. 6-Lead Limb)

I ACTIVE	II ACTIVE	III ACTIVE	aVR ACTIVE	aVL ACTIVE	aVF ACTIVE
V1 MASKED	V2 MASKED	V3 MASKED	V4 MASKED	V5 MASKED	V6 MASKED

[VERIFIED] INPUT PROVENANCE AUDIT PASSED

- Stratified patient split across PTB-XL, Chapman, and MIMIC-IV confirmed.
- Zero data leakage between train, calibration, and quarantined test sets.
- Hardware lead mask cleanly zeroes missing leads without synthetic artifacts.

PART B: REASONING TRACE AUDIT & THE LIMITATION

Cross-Attention Weight Distribution on Degraded Input

Model predicts 'Myocardial Infarction' (Score = 0.867) under 6-lead telemetry



LIMITATION **UNGROUNDED REASONING COLLAPSE**

1. THE BLIND HALLUCINATION PROBLEM:
   While classification AUROC stays deceptively high (~0.95) down to 2 leads, the model's reasoning transformer attributes its diagnosis to disconnected chest leads (V1, V4) that are 100% physically absent from the patient.

2. ACCURACY CONCEALS SAFETY FAILURES:
   Standard diagnostic benchmarks award full credit to these predictions, blind to the fact that the underlying reasoning trace is completely ungrounded.

3. REQUIRED SCIENTIFIC GUARDRAIL:
   Stronger clinical conclusions cannot be drawn until evidence-grounding checks are established to audit reasoning integrity against hardware masks.